A Search for Laser Lines in Five Large Spectroscopic Surveys of Nearby Stars

Charles C. Norton

Abstract

A laser aimed at Earth from another star adds an unresolved emission line to the star's spectrum. The large spectroscopic surveys now hold spectra of tens of thousands of stars within 100 pc, taken for stellar parameters and abundances, and none has been searched for such lines. We search five: the Gaia DR3 mean RVS spectra of 2363 FGK stars within 75 pc (846 to 870 nm), GALAH DR4 spectra of 621 dwarfs within 100 pc (471 to 789 nm in four windows), APOGEE DR17 combined spectra and 14178 visits of 3370 dwarfs within 100 pc (1.51 to 1.70 μm, including the 1530 to 1565 nm band of terrestrial fibre-optic lasers), DESI DR1 spectra of 7705 stars within 100 pc (360 to 982 nm), and LAMOST DR10 medium-resolution spectra of 14400 stars within 100 pc in 35179 observations. Each spectrum is modelled by a non-negative combination of the most similar spectra of its survey, so a narrow emission line that they lack cannot be fitted away, and emission-like residuals are judged against the absorption-like residuals of the same spectra. No emission line above that null remains in the Gaia, APOGEE and multi-exposure DESI spectra; every GALAH candidate is instrumental; and in LAMOST the 19 features that recur in two exposures of single stars are one or two pixels wide where the instrument's profile spans six, detector pixels and not light. From each spectrum's noise and the star's magnitude, a line at the detection threshold carries a median of 2×10^{-16} erg cm^{-2} s^{-1} in APOGEE and DESI and 1 to 3×10^{-15} in GALAH, Gaia and LAMOST, the flux of a transmitter of 10^{19} to 10^{20} W radiating isotropically or of 5 kW to 70 kW behind a 10 m aperture at the median star, and of 5 to 50 W at the nearest.

1 Introduction

Schwartz and Townes proposed that a civilization could signal with optical masers, whose light, confined to a narrow beam and a narrow band, would outshine its star at that wavelength [1]. A laser aimed at Earth would add an unresolved emission line to the star's spectrum; from a 10 m aperture the beam is a few hundredths of an astronomical unit wide at 10 pc and about one at 100 pc, so it has to be held on the Earth's position when the light arrives. Searches have looked for such lines in the high-resolution spectra of radial-velocity programmes: 577 FGKM stars with Keck/HIRES [2], 2796 stars at 3640 to 7890 Å [3], 5600 FGKM stars with thresholds of megawatts [4], 388 stars with the Automated Planet Finder [5], and 2821 stars in archival HARPS spectra, where three candidates in two M dwarfs and a red giant remain to be explained by the stars themselves [6].

The spectroscopic surveys of the last decade observe many more nearby stars, at lower resolution and over different wavelengths: the Gaia RVS spectra near the calcium triplet, GALAH's four optical windows, APOGEE's H band, which contains the 1530 to 1565 nm band in which Earth's fibre-optic lasers operate, and the wide optical coverage of DESI and LAMOST. Their spectra are taken in many fibres at once and reduced in bulk, so a search must tell a line in a star from the defects of fibres, detectors and sky subtraction. This paper searches the nearby stars of all five surveys with methods that compare each spectrum with others of the same survey.

1.1 Contributions

1. A search for unresolved emission lines in the spectra of nearby stars from Gaia DR3, GALAH DR4, APOGEE DR17, DESI DR1 and LAMOST DR10, about 28000 stars counted survey by survey.
2. Comparison-spectrum models in which a narrow line that the comparison stars lack cannot be fitted away, with each spectrum's own absorption-like residuals as the null.
3. The instrumental and terrestrial features of each survey that mimic a line, each turned into a test.
4. The line flux at each survey's threshold, star by star, from the spectrum's noise and the star's Gaia or 2MASS magnitude, as an isotropic luminosity and as the power of a 10 m transmitter aimed at Earth.

2 Data and method

The common method. Each spectrum is divided by its upper envelope or continuum and fitted, window by window, as a non-negative combination of the spectra of its most similar stars in the same survey, with a signed continuum and signed first and second derivatives of the fitted model, which absorb small differences of velocity and line width. A narrow emission line in the target cannot be reproduced unless a comparison spectrum carries it. The residual is searched with a matched filter of the instrument's line profile, and peaks of both signs are kept: the absorption-like peaks arise from the same imperfections of the model and the data as false emission-like ones, and serve as their null. A peak of unresolved width is then tested against the same position in other stars, in the stellar rest frame (a stellar line the model misses) and in the observatory frame (sky, telluric and instrumental features).

Gaia RVS. The mean RVS spectra of Gaia DR3 [7, 8], 846 to 870 nm at a resolving power of 11500, of the 2363 FGK stars within 75 pc that have one, are each fitted by their 25 nearest neighbours in colour and absolute magnitude. A positive peak counts only above every absorption-like peak of its environment, and is tested against the cores of the Ca II triplet and the Paschen lines, the same position in other stars, and deblended neighbours.

GALAH. GALAH DR4 [9] gives 699 spectra of 621 dwarfs within 100 pc, at a resolving power of 28000 in four windows from 471 to 789 nm. Each spectrum, camera by camera, is fitted by the 20 sample spectra nearest in effective temperature, gravity and metallicity; a candidate is then measured in the other fibres of the same exposure.

APOGEE. APOGEE DR17 [10, 11] holds H-band spectra at a resolving power of 22500. For the 3370 dwarfs within 100 pc with three or more visits, the combined spectrum is fitted in 256-pixel windows by its 12 most similar stars, and peaks under bright subtracted sky are OH airglow. Separately, each of the 14178 visits is divided by the star's median visit, which cancels anything present in every visit, and lines that recur in several visits at one rest wavelength but at different observatory wavelengths are grouped; detector dropouts that depress the median are masked.

DESI. The DESI DR1 [12] Milky Way Survey [13] spectra of 7718 stars with Gaia parallaxes of 10 to 800 mas and S/N above 20, 6266 of them cooler than 3500 K and 455 white dwarfs, are fitted by their 12 most similar spectra; each candidate is checked exposure by exposure, since a coadd of several exposures can carry a cosmic ray or a defect of one of them.

LAMOST. LAMOST DR10's medium-resolution survey [14, 15] observes 495 to 535 and 630 to 680 nm at a resolving power of 7500 in several exposures a night. In the 35179 observations of 14400 Gaia DR3 stars within 100 pc with declination above $-15°$ and $G < 15$, a peak counts at 6σ above

Table 1: The surveys searched and their results. The null is the set of absorption-like residual peaks found by the same method in the same spectra.

Survey	Range	R	Stars	Result
Gaia DR3 RVS	846–870 nm	11500	2363	no line above the null
GALAH DR4	471–789 nm	28000	621	every candidate instrumental
APOGEE DR17, combined	1.51–1.70 μm	22500	3370	no line above the null; one peak above it is OH airglow
APOGEE DR17, visits	1.51–1.70 μm	22500	3370	3431 recurring emission-like groups against 3410 absorption-like
DESI DR1	360–982 nm	2000–5000	7705	coadds of two or more exposures match the null
LAMOST DR10 MRS	495–535, 630–680 nm	7500	14400	19 two-exposure features, all detector pixels

the local continuum with an unresolved width in a single exposure, and persists when two exposures hold it within a third of a resolution element in the star's frame. A feature is explained by an arm's end, by sharing with other stars in the stellar or the instrument's frame, by recurrence in the same or neighbouring fibres of a spectrograph, by instrument-frame recurrence on other nights, or by light in neighbouring fibres of the same exposure, and what remains is set against the coincidences of two exposures that chance makes and read again at its peak pixel in every exposure of its star: a feature narrower than half the instrumental profile is a detector pixel, one whose exposures peak more than half a resolution element apart is not one feature, and one held by a single exposure of a night whose other exposures are flat is a cosmic ray.

Flux limits. Each search's threshold is a matched-filter significance k in a residual that is in units of the continuum: $k = 5$ for the Gaia, APOGEE and DESI searches, 5.5 for GALAH and 6 for a LAMOST exposure. An unresolved line whose filter response sits at the threshold has a peak of $k\sigma/g$ of the continuum, with σ the spectrum's noise per pixel and g the filter's gain, and a flux $F = (k\sigma/g)\sqrt{2\pi}\,\sigma_\lambda F_\lambda$, with σ_λ the Gaussian width of the line profile and F_λ the star's continuum flux density, which we take from its Gaia BP, G and RP magnitudes, interpolated in wavelength in the Vega system, and from its 2MASS H magnitude for APOGEE. For Gaia the noise is that of each star's residual; for GALAH, APOGEE and DESI it is the survey's per-pixel signal-to-noise ratio of each camera or of the combined spectrum; for LAMOST it is the per-exposure ratio of each arm, read from the headers of a random 1500 of the 35,179 observations. An isotropic source at the star's Gaia distance d then has $L = 4\pi d^2 F$, and a transmitter aimed at Earth through a diffraction-limited 10 m aperture needs $P = F d^2 (\lambda/10\,\mathrm{m})^2$. These are thresholds, not completeness limits: in the ultraviolet search of the same kind, lines injected into archival spectra were recovered at 90 per cent of the attainable fraction at 3 to 5 times the nominal threshold, and the same factor applies here, since the same envelopes, masks and comparison fits intervene.

3 Results

Table 1 summarises the results.

Gaia RVS and APOGEE. No emission-like peak in the RVS spectra rises above the absorption-like null. In the APOGEE combined spectra no emission-like peak exceeds the null either in the continuum or inside stellar lines, once OH airglow is masked where the visits' subtracted sky was bright; the one peak above the null was such airglow. In the visit-to-visit search the recurring

emission-like groups number 3431 against 3410 absorption-like ones, and the strongest, 48.2, falls below the strongest absorption-like group, 67.0.

GALAH. Every GALAH candidate is instrumental: pixels that several stars share in the observatory frame, charge persistence in one fibre from a bright earlier exposure, cosmic rays, and a ghost in the blue camera.

DESI. The 1091 stars with three or more exposures match the absorption-like null in the continuum. Every peak above the null in them, and in the 1878 coadds of two exposures, vetted exposure by exposure, is a cosmic ray in one exposure, a corrupt exposure, an OH residual, or a mismatch of the comparison spectra in a telluric band or in a white dwarf's Paschen lines. The 4749 stars with a single exposure, 62 per cent of the sample, have no second exposure to separate a line from a cosmic ray, so for them the search ends at the peak list: their emission-like peaks are as numerous as their absorption-like ones, and they are not searched in the sense the others are.

LAMOST. The single exposures hold 12261 unresolved spikes at 6σ, and 110 persist in two exposures in the star's frame. Arm ends, lines shared in the stellar or the instrument's frame (sky residuals moved by the heliocentric correction), defects of the same or neighbouring fibres, recurrence in the instrument's frame on other nights and light in neighbouring fibres of the same exposures explain 91; the 19 left (Table 3) lie in single stars, mostly in two of many exposures. The strongest, at 241σ in three exposures of one night of a K dwarf, lies a few pixels from a block of pixels at 1800 times the continuum in all five exposures of that night, with the pipeline's variance there inflated a thousandfold: a defect of those extractions, absent from the star's 178 other exposures, against which the residual's scatter was measured. The other 18 are as many as the 17 coincidences of two exposures that chance makes, and read at the peak pixel in every exposure of their stars, all 19 are detector pixels: each rises above half its excess over one or two pixels, where the instrumental profile's full width at half maximum is 5.6 to 6.5 pixels, so no light passing through the spectrograph made it. Six of them sit as well on masked, variance-inflated or dead pixels, eight have exposures whose peaks lie more than half a resolution element apart, four fill one night and none of the star's other nights, and three are single exposures of nights whose other exposures are flat.

Flux limits. Table 2 gives the line flux at each survey's threshold. The near-infrared and the deep DESI coadds reach $2\times10^{-16}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$ at the median star and the optical surveys of brighter stars 1 to 3×10^{-15}; as isotropic luminosities at the stars' distances these are 10^{19} to $2\times10^{20}\,\mathrm{W}$, and as transmitters behind a 10 m aperture aimed at Earth 5 to 70 kW at the median star, 300 W at the DESI stars within 20 pc and 5 to 50 W at the nearest star of each survey. The thresholds of Tellis and Marcy's search, megawatts for a 10 m aperture at their stars' distances [4], are reached here at every survey's median star, with the three-to-five-fold completeness factor of the previous paragraph still to be applied.

4 Related work

Laser searches. The searches of Section 1 use high-resolution spectra of radial-velocity programmes [2, 3, 4, 5, 6], and the HARPS search, the largest of them, ends as this one does with a handful of features in stars whose own spectra are the first thing to exclude. We find no earlier laser search of Gaia, GALAH, APOGEE, DESI or LAMOST spectra.

Table 2: The line flux at each search's threshold: the median over the stars (for GALAH, DESI and LAMOST over the cameras, arms and exposures) of F, of the isotropic luminosity L at the star's distance and of the power $P_{10\,\mathrm{m}}$ of a 10 m transmitter aimed at Earth; the number of stars within 20 pc; and $P_{10\,\mathrm{m}}$ at the nearest star. The LAMOST row is from the 1163 stars among a random 1500 observations whose headers were read. APOGEE's $P_{10\,\mathrm{m}}$ is larger than DESI's at a smaller flux because the beam at 1.6 μm is five times wider.

Survey	Stars	F (erg cm^{-2} s^{-1})	L (W)	$P_{10\,\mathrm{m}}$ (W)	$N_{<20\,\mathrm{pc}}$	$P_{10\,\mathrm{m}}$, nearest (W)
Gaia DR3 RVS	2363	2.9×10^{-15}	1.2×10^{20}	7×10^{4}	27	3.5×10^{4} (7.2 pc)
GALAH DR4	621	1.3×10^{-15}	1.1×10^{20}	3×10^{4}	4	50 (4.3 pc)
APOGEE DR17	2931	1.6×10^{-16}	1.1×10^{19}	2×10^{4}	4	2.5×10^{3} (13.8 pc)
DESI DR1	7718	2.1×10^{-16}	1.4×10^{19}	5×10^{3}	96	5 (3.6 pc)
LAMOST DR10 MRS	1163	3.0×10^{-15}	2.0×10^{20}	5×10^{4}	20	4×10^{3} (16 pc)

5 Conclusion

No unresolved emission line that a laser aimed at Earth could produce is found in the nearby stars of five large spectroscopic surveys, beyond what the surveys' instruments, the sky and chance produce. The surveys extend the earlier laser searches to tens of thousands of stars, to the near infrared and to the telecommunications band.

The limits are thresholds in each spectrum's noise, turned into fluxes by the stars' magnitudes, and apply to lines that are steady in the combined spectra or that recur in the exposures available: 10^{19} to 10^{20} W isotropic at the median star, kilowatts to tens of kilowatts behind a 10 m aperture, watts at the nearest stars. A line inside a strong stellar line, one in the telluric bands that the models fit away, or one that was off when the spectra were taken is not constrained; the 62 per cent of DESI stars with a single exposure need a second, and injecting lines into the archival spectra, as the ultraviolet search did, would turn these thresholds into completeness limits survey by survey.

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Table 3: The 19 LAMOST features that pass the tests against other stars, fibres and nights: the star, the arm, the rest and observed wavelengths (air), the exposures holding the feature against the star's exposures, the nights they span, and the largest single-exposure significance. Each is one or two pixels wide.

Gaia DR3	Arm	λ_{rest} (Å)	λ_{obs} (Å)	Exposures	Nights	z
4420231361513433344	B	4947.67	4947.83	2/6	1	9.5
1407678284339274368	B	5102.09	5102.55	2/8	1	9.4
1407678284339274368	B	5134.99	5135.46	2/8	1	8.3
608182831000585728	B	5269.81	5270.60	2/30	1	7.8
815877833000303488	B	5290.17	5289.98	2/238	2	9.1
1295249440828061696	B	5331.15	5331.32	3/160	1	20.0
1894835705805757824	B	5344.89	5345.21	2/152	2	17.8
64953069592851328	R	6307.36	6306.18	2/124	2	11.1
1898531439263850240	R	6322.92	6323.26	5/106	1	14.5
814125692502763904	R	6335.08	6335.53	2/214	2	22.4
687208343753227520	R	6433.52	6433.98	2/52	2	28.2
1547717514092903424	R	6551.41	6550.29	2/196	2	9.9
2556424771496343424	R	6626.80	6626.62	2/76	2	9.0
1915323176544379520	R	6728.65	6729.16	2/180	2	11.7
779262652684989312	R	6756.41	6756.55	2/174	2	6.5
3872246689058776832	R	6764.64	6774.04	2/123	2	33.0
1386361674454950912	R	6772.31	6772.58	3/183	1	241.4
161252970877957632	R	6783.43	6783.15	4/34	1	7.0
986071444021690112	R	6801.65	6802.17	4/22	1	21.1

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